Reading a Month of Attack Traffic: A Power BI Analysis of 148,580 Honeypot Connections

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Data*: one T-Pot/Cowrie honeypot sensor, Sept 3 to Oct 3, 2026 (UTC). All IP addresses in this report are masked to protect the owners of those addresses.*

At a glance: Built in Power BI (Power Query, DAX, data modeling) from 31 daily log files: about 483,000 events, 148,580 connections and 7,103 distinct IPs. Deliverables: a four-page dashboard and this report.

Summary

This report analyzes 31 days of connection logs from a single honeypot, a decoy server that accepts SSH and Telnet logins so that automated attackers reveal what they try. I turned roughly 483,000 log events into a four-page Power BI dashboard built around four questions: when activity happens, where it comes from, how sources compare with each other, and what attackers do once they get in. I did not set out to judge the sensor or its security. The aim was to find meaningful trends and to show how I got to them.

The main findings:

- **Volume is driven by a few sources, not by many.** The sensor logged 148,580 connections from 7,103 distinct IP addresses. The largest daily spike (Sept 6, 14,258 connections) came mostly from a single address that ran at a steady rate for about four hours. Other busy days look the opposite: hundreds of distinct IPs, each making only a few connections.
- **Two address blocks account for 38% of all connections.** One block of 256 addresses (a /24) located in Bulgaria produced 26.2% of all connections from 79 IPs, and a block located in the US produced 11.8% from 43 IPs. Together, 122 IPs (1.7% of sources) made 38% of the traffic.
- **Few sessions get in, and those that do follow a script.** About 77% of connections try a login and 4.3% of those succeed. Of the 4,901 successful logins, 72% run commands, mostly system reconnaissance plus what looks like an SSH-key installation routine. Only 61 distinct files were ever captured, and one of them accounts for 79% of file-capture sessions.
- **Usernames are concentrated, passwords are not.** The username 'root' alone is 38% of login attempts, while the ten most common passwords together are only 16%.

- **Protocol choice varies sharply by country.** Traffic from Pakistan is 92% Telnet, while traffic from Bulgaria, Poland and Romania is more than 99.7% SSH.

The lesson that ran through the whole project is methodological: headline totals and averages repeatedly hid the real structure. Looking at distributions, and always showing volume next to the number of distinct sources, changed the interpretation almost every time.

1. Background and Goal

A honeypot is a deliberately exposed system that exists to be attacked, so that the attempts can be observed safely. T-Pot is a platform that bundles many honeypots. I worked with its Cowrie component, which imitates SSH and Telnet servers: it accepts logins, records every command typed, and captures any files that attackers send or fetch.

A friend who operated the sensor shared the logs. He wrote his own threat-research report from the same data, so this report is independent and has a different purpose: it is analytical, aimed at trends and at the data-handling work needed to find them. I chose to build it in Power BI because that was a skills gap I wanted to close.

My goal was to show that I can take a large, messy, real-world dataset from raw files to a tested dashboard and a written analysis, and to document my decisions, questions and mistakes along the way. I organized the work around four angles: time series (when), geography (where from), comparison (who, and how they differ) and post-compromise behavior (what happens after a login).

2. The Data

- **Format:** 31 daily files in JSON Lines (one JSON object per line), compressed with gzip apart from the newest day. The logs are an event stream: each connection produces a sequence of events (connect, login attempts, commands, close), and the fields differ by event type.
- **Window:** Sept 3 00:00 to Oct 3 10:00 UTC. The first day is complete; the last is a partial day.
- **Scope:** the T-Pot export contained many honeypot types. I used Cowrie only, because it is the richest and most interpretable and is enough to answer all four questions.
- **Integrity and safety:** I verified the download against its SHA-256 checksum (hex digests are case-insensitive, so upper- and lower-case versions match). The archive was treated as potentially hostile: nothing executable was run, no links were opened, and captured files were analyzed only by hash and count.
- **Privacy:** at the request of the sensor's operator, IP addresses are masked. Single IPs show the last number as 'xxx', address ranges appear as /24 blocks, and the dashboard's only IP-level table is masked. The raw logs are not shared.

3. Preparing the Data

I combined the 31 files with Power Query, filtered to the events I needed, and kept every transformation as a documented step. The combining was the easy part. These five problems were the real work:

- **A load that hung for over an hour.** SSH client-fingerprint events carry large nested arrays that Power Query parsed before filtering. Those events do not feed any of my four questions, so I excluded them early and the load completed. A leftover auto-generated column then errored on every row; removing it fixed that.
- **Timestamps shifted six hours.** A partial Sept 2 appeared even though collection starts Sept 3 UTC. A check on the earliest timestamp confirmed Power Query had converted UTC to local time on load, so I added a UTC column and rebuilt the date from it. The constant offset is valid because the whole window falls inside daylight time.
- **Geolocation needed a range match.** The country lookup maps blocks of addresses, so a plain join fails. I converted addresses to numbers, geolocated only the distinct IPs (thousands of rows, not hundreds of thousands), and linked the result to the events with a relationship, because a merge made Power Query re-run the whole pipeline on every preview.
- **One column meant two things.** Of 1,689 distinct file hashes, 1,629 were hashes of recorded terminal sessions, not files. Restricting download and upload events gave the real figure: 61 distinct files.
- **Shortcuts and averages gave wrong or misleading numbers.** A visual's built-in 'percent of grand total' divided by the visible rows only, putting Bulgaria at 36% of connections instead of 27.4%, so I used explicit measures and reconciled them against the tables. Poland's 389 connections per IP was a single address, so I added a 'top IP share' measure beside every average.

I also scoped the columns deliberately: of 29 columns I kept 15, keeping a column only if it fed one of the four questions and had more than one distinct value. Each removal has a documented reason.

4. Dashboard Design

The dashboard has four pages, one per question: Time Series, Geolocation, Comparative Analysis and Post-Compromise. I followed four rules:

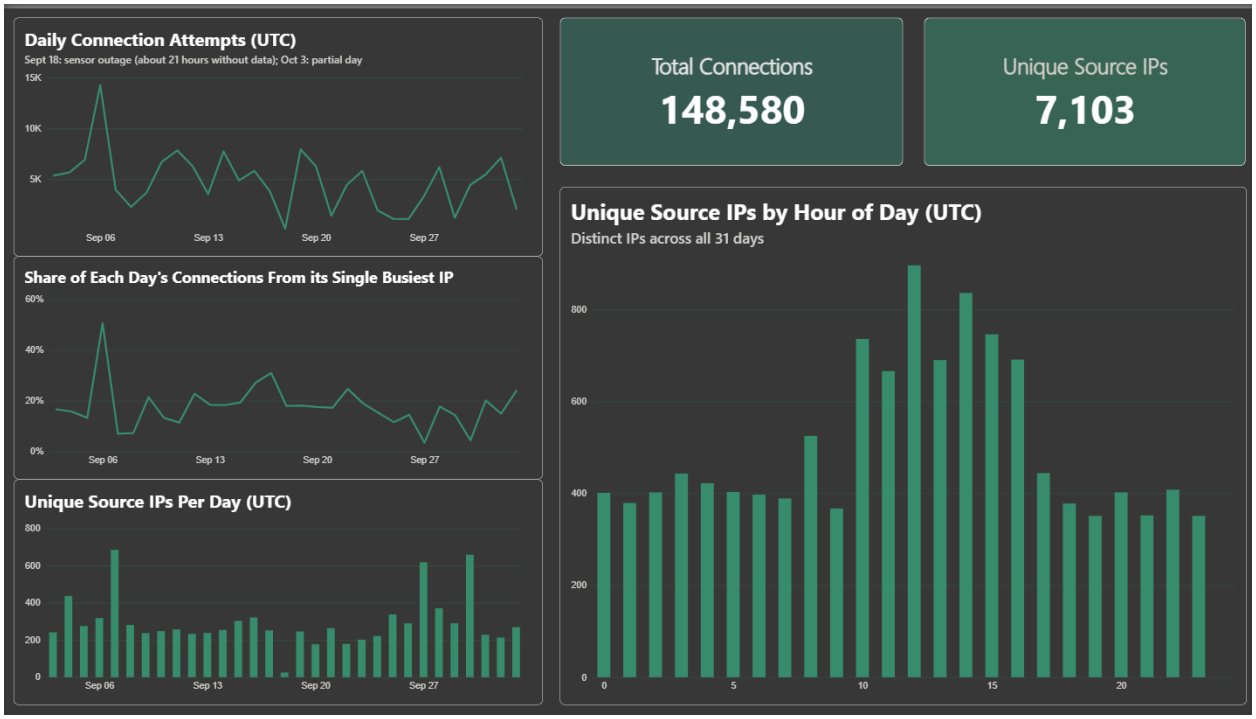
- **Volume is always shown next to breadth** (distinct IPs), so one heavy IP cannot pass for a trend.
- **The same color always means the same measure,** and every title says what is measured and in which time zone.
- **Every number reconciles with another view:** funnel stages equal the card values, and shares add up to their totals.

- **Investigation views stay off the dashboard.** Exploratory tables live on a hidden scratch page, and the findings are written up here.

Technically, the model is an events table plus a small distinct-IP table linked by a relationship, with all calculations written as measures in DAX (see the Appendix).

5. Findings

5.1 When: spikes come in two kinds



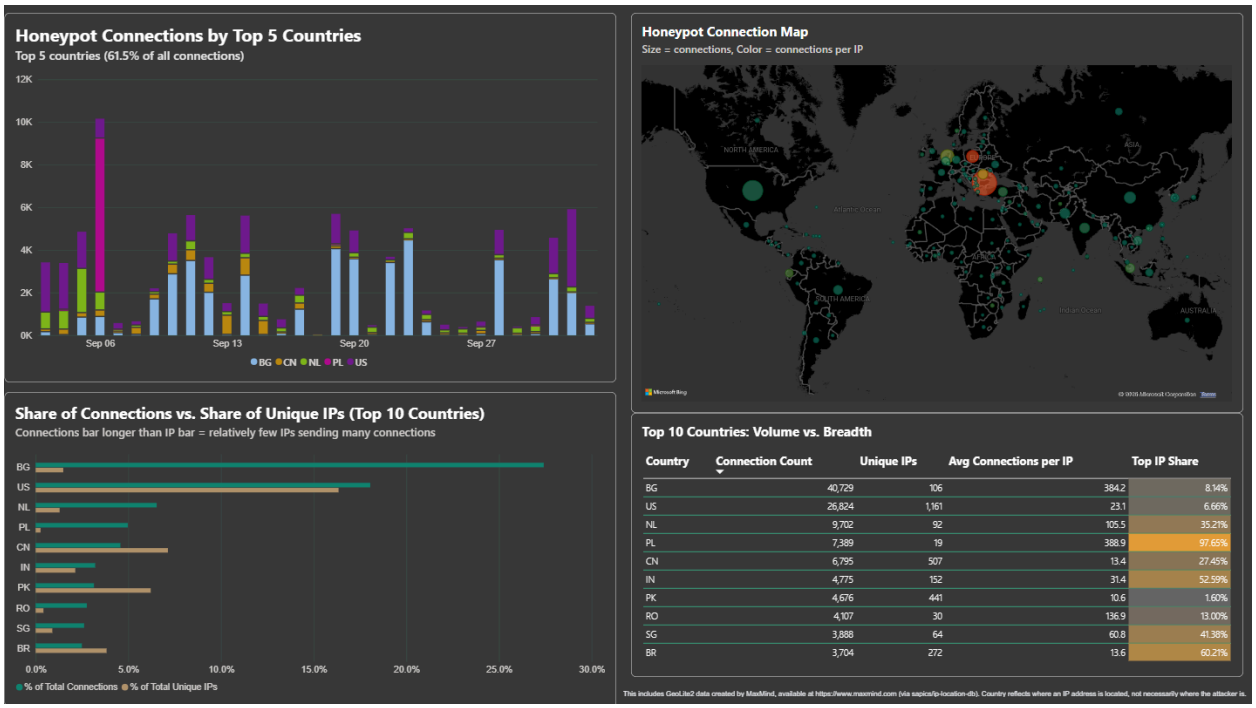
Most days see between roughly 1,000 and 8,000 connections (about 4,800 on average). The largest spike, Sept 6, reached 14,258 connections from 318 distinct IPs. A single address in Poland made 7,215 of them, 51% of the day, and it appears on no other day. Its activity fits inside roughly four hours at a nearly constant rate of about 1,800 connections an hour, one every two seconds, which points to an automated script.

Distinct IPs per day tell a different story. On most days between about 180 and 370 distinct addresses connect, but a few days (one has roughly 685 IPs against only about 4,000 connections) show many more sources at ordinary volume. So there are two kinds of spikes: bursts, where a few IPs make a great many connections, and sweeps, where many IPs make a few each. Plotting the daily share of connections from the busiest IP separates them: about 51% on the burst day, and typically 10 to 30% otherwise.

Across the day, connection volume is fairly flat around the clock, but distinct IPs more than double between 09:00 and 12:00 UTC (367 to 896). The hourly volume view is also skewed: the

single Poland burst adds about a quarter to each of three hours, which is why I use distinct IPs for the hour-of-day chart.

5.2 Where from: a few address blocks dominate



The ten largest countries account for 75.8% of connections but only 40% of distinct IPs. Bulgaria alone produced 40,729 connections (27.4%) from 106 IPs (1.5% of sources), while the US produced 26,824 from 1,161 IPs.

Averages mislead here. Poland's high connections-per-IP figure is one address (98% of the country's traffic), and Brazil (60%) and India (53%) are also more than half one IP. Bulgaria is different: its busiest IP is only 8% of its traffic, so its volume comes from roughly a hundred IPs each working hard.

A block-level view explains why. One /24 block in Bulgaria made 38,949 connections, 26.2% of everything and about 96% of Bulgaria's traffic, from 79 IPs active on 28 of 31 days. A US block made 17,537 connections from 43 IPs on 30 days, about two-thirds of US traffic. I had first read the US as broad background activity, and only the block view corrected that. The sources fall into three patterns: always-on fleets (these two blocks), single-host bursts (the Polish address) and wide, shallow background scanning (for example Pakistan and China, with hundreds of IPs each making a few connections).

5.3 How sources compare: credentials, protocols and top IPs

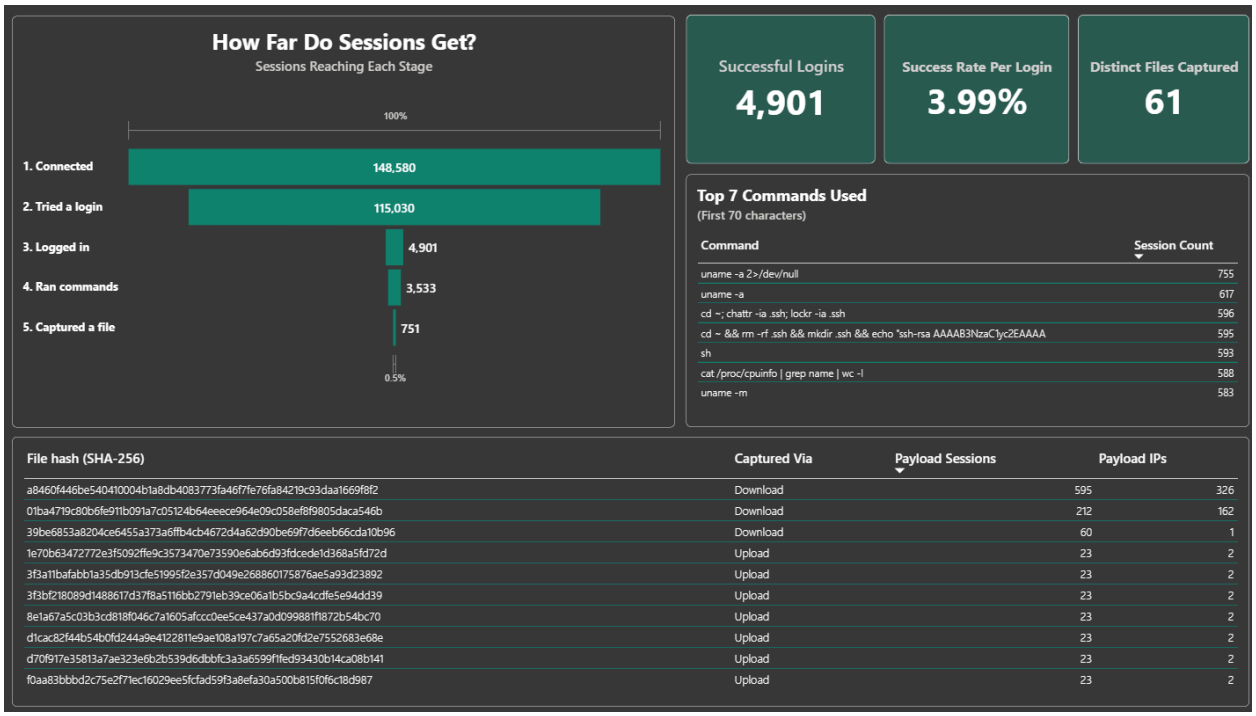


Credentials. The username 'root' is 38.4% of about 123,000 login attempts (47,179 attempts from 1,711 IPs), and the ten most common usernames cover about 55%. Passwords are far more spread out: the top ten cover about 16%, led by 123456, 123 and 1234. Each is tried from roughly 190 to 590 distinct IPs at only a few attempts per IP, the signature of widely shared password lists, not of one operator working through its own.

Protocol. Only ports 22 (SSH) and 23 (Telnet) are targeted. The mix is a country-level trait: 92% of Pakistan's connections are Telnet (spread over 441 IPs, no dominant address), as are 93% of Belgium's, while Bulgaria, Poland and Romania are more than 99.7% SSH.

Top source IPs. The ten busiest IPs (0.14% of sources) made 20.4% of all connections. They are a mix: a Netherlands host active on all 31 days at about 110 connections a day, hosts in India and Turkey active on 19 and 16 days, and burst-and-leave addresses such as the Polish IP (one day) and three addresses in the Bulgarian block (two to four days each, at 630 to 1,200 connections a day).

5.4 What happens after login



Of 148,580 connections, 115,030 (77%) try a login, 4,901 sessions get in (4.3% of those; 3.99% of individual attempts), 3,533 (72% of successes) run commands, and 751 (21% of those) capture a file. End to end, 0.5% of connections end with a file captured.

The commands are system reconnaissance followed by what looks like an SSH-key persistence routine: delete the .ssh folder and install the attacker's own key. Five commands appear in 583 to 596 sessions each, consistent with one script run start to finish. Only 61 distinct files were captured (21 downloaded, 40 uploaded), and the most common one appears in 595 of the 751 file-capture sessions (79%), probably the same sessions as the key-installation command. Seven uploaded files each appear in exactly 23 sessions from the same two IPs, which looks like one toolkit uploaded as a bundle. I have not confirmed either link by opening individual sessions, so I treat both as hypotheses.

6. Putting It Together

The traffic falls into three kinds: always-on fleets that work a block of addresses almost every day, single-host bursts that appear for hours or one day, and a wide, shallow layer of background scanning from thousands of IPs. Raw connection counts blend all three, so a count alone says little about how many parties are involved or how persistent they are. Reporting volume next to distinct sources, and measuring at the block level as well as the IP level, separates them. The narrow funnel is consistent across them: credential guessing at scale, a small number of successful logins, and a scripted routine once inside.

7. Limitations

- **One sensor, one honeypot type, one month.** Patterns may reflect where this sensor sits and should not be generalized to the internet. Login success also depends on how the honeypot is configured, so the 4% success rate describes this sensor, not real servers.
- **Geolocation is approximate, and a block is not an actor.** Country reflects where an address is registered or located, not where an attacker is, and hosting providers can rent addresses in one block to different customers, so the 'fleet' reading is a hypothesis.
- **Distinct hashes are not malware families,** so 61 counts files, not kinds of malware.
- **Data gaps.** The Sept 18 sensor outage (about 21 hours) and the partial Oct 3 understate those dates, and the six-hour timezone correction assumes the whole window is inside daylight time.
- **Untested ideas.** I did not test whether repeated exact connection counts (for example several IPs at exactly 1,099) reflect fixed-size scripted runs, and I did not confirm the key-installation and toolkit-bundle links by opening individual sessions.

8. What I Learned

- **Check the denominator and the distribution.** Averages, totals and built-in percentages misled me at least four times, and each correction changed a conclusion.
- **Validate what a field means before building on it.** The file-hash column held two different things, and it would have inflated the payload count by a factor of about 28.
- **Design the model to avoid expensive rework.** Moving from a merge to a relationship, and filtering before parsing, removed repeated hour-long waits.
- **Document decisions as you go.** I kept a running log of steps, questions and corrections, which is how I could trace and fix errors like the time zone shift.

Skills demonstrated: Power Query (JSON Lines, folder combine, early filtering), data modeling, DAX measures, validation and reconciliation, dashboard design, and safe, anonymized handling of security data.

Appendix: Key Measures

- **Connection Count:** connection events (one per session).
- **Unique Source IPs:** distinct IPs that connected.
- **Connections per IP:** connections divided by unique IPs; always read together with Top IP Share.
- **Top IP Share:** the share of a group's connections made by its single busiest IP.
- **Share of Connections / Share of Unique IPs:** a group's share of all connections and of all distinct IPs.
- **Login Attempts / Successful Logins:** login events, failed plus successful, and successful only.
- **Active Days:** distinct UTC days on which an IP made at least one connection.
- **Sessions per funnel stage:** distinct sessions that connected, tried a login, logged in, ran commands, and captured a file.
- **Distinct Files Captured:** distinct file hashes from download and upload events only.
- **Subnet (/24) and IP Label:** the first three numbers of an address plus '.0/24', and a masked label that keeps the last number hidden.